

THE STORY OF MEDICINE

ANCIENT BELIEFS GIVE WAY TO SCIENTIFIC ADVANCES, TRANSFORMING HUMAN HEALTH

Understanding of the human body and disease made important advances during the 18th and 19th centuries, laying the foundation for modern medical care. Ancient practices, such as bloodletting to cure illness, were replaced with ones that were born of a more rigorous scientific approach.

People have attempted to treat disease since prehistoric times, but until the 18th century medicine was based largely on superstition, natural remedies, and unscientific practices and theories, such as the theory that the body had four fluids (humors) that needed to be in balance for health. There had been progress in anatomy and surgery, but overall, medicine remained primitive.

THE DEVELOPMENT OF MODERN MEDICINE

In the 18th century, medicine started to become more scientifically rigorous, and significant advances were made, such as the development of a vaccine for smallpox in 1796. The 19th century saw the establishment of the germ theory of disease,

the introduction of antiseptic techniques and anesthetics, and the use of X-rays to image the body. Around 1900, pharmacology began to make great progress, with the invention of aspirin in 1897 and the first synthetic antibacterial drug in 1908. During the 20th century, more vaccines and drugs were developed, such as antibiotics and anticancer drugs. Surgical techniques also became more sophisticated; successful organ transplants were performed, and keyhole surgery became routine. In diagnosis, scanning techniques were invented, and screening became widely used. From the late 20th century, genetics also began to have a significant impact on medicine as genetic causes of diseases were discovered and genetic testing was developed.



THE STETHOSCOPE

The stethoscope was invented in 1816 by French physician René Laennec, who used a simple tube (a monaural stethoscope) to listen to a woman's chest. In 1851, British physician Arthur Leared invented the binaural stethoscope, with an earpiece for each ear, and, in the 1940s, Americans Maurice Rappaport and Howard Sprague developed the modern acoustic stethoscope, which has two "bells," one for listening to the heart, the other for listening to the lungs. The latest development is the electronic stethoscope, which uses an electronic sound sensor and amplifier.

“THE DEVIATION OF MAN FROM THE STATE IN WHICH HE WAS ORIGINALLY PLACED BY NATURE SEEMS TO HAVE PROVED TO HIM A PROLIFIC SOURCE OF DISEASES.”

Edward Jenner, English surgeon, *An Inquiry into the Causes and Effects of the Variolae Vaccinae, or Cow-Pox*, 1798

c. 5100–4900 BCE

Neolithic trepanation
Trepanation, which involves drilling holes in the skull, is used as far back as the Neolithic period to treat a variety of health problems.



Trepanned skull

c. 420 BCE

Hippocrates develops diagnostics
Hippocrates, the Greek physician considered to be the father of modern medicine, moves health away from religion and into the realm of science.

1543

Andreas Vesalius
The Brussels-born surgeon writes his influential anatomical work, with accurate diagrams of human anatomy based on many dissections and operations.



De Humani Corporis Fabrica

1818

First blood transfusion
British obstetrician James Blundell performs the first successful human-to-human blood transfusion, using a syringe to transfer blood between the patients.

c. 1550–700 BCE

Ancient Egyptian surgery
Medical and especially surgical knowledge advances due to the practice of mummification, which gives doctors greater insight into anatomy.



Egyptian knives and curettes

c. 1000–1300

Arab medical advances
The Arab world adds to medical progress with the development of pharmacists, who work with plants and use them to find new cures.



Arabic medical manuscript

1796

Vaccination
British scientist Edward Jenner develops a vaccine for smallpox. It is the first vaccine created for any disease, and Jenner's work saves countless lives.



Jenner's inoculation point